

AdvPrint Pro

Advanced Printer — silent printing for every web page, PDF, application and receipt printer

by Bit Dream IT — Innovative Healthcare Solutions

Version 3.0.1 — Enterprise Edition

Full Setup Guide — A to Z

100% local · no cloud · no tracking · MIT licensed

Windows 7/8/10/11 · Windows Server · Debian/Ubuntu/Mint · macOS

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1. What AdvPrint Pro is — and how one print job travels

AdvPrint Pro (Advanced Printer) by **Bit Dream IT — Innovative Healthcare Solutions** prints anything — PDFs, web pages, `window.print()` calls, receipts (raw/ESC-POS), screenshots, selections — **silently**, with no print dialog, on any printer connected to the hub PC, including printers shared from other PCs. Everything stays on your machines: the hub listens on `127.0.0.1:9235` only (until you deliberately enable LAN sharing), no cloud, no telemetry.

> Product identity: **AdvPrint Pro** — *AdvPrint (Advanced Printer)* — **by Bit Dream IT — Innovative Healthcare Solutions**. The hub reports `app: "AdvPrint Pro"` and the full brand string in every /health answer.

```
Chrome (any web page)
   │  extension: auto-print rules / right-click / Alt+P / window.advprint API
   ▼
AdvPrint Pro HUB  (http://127.0.0.1:9235, token-protected)
   │  queue → retries → holds → printer status checks
   ▼
Silent print engines:  Windows: SumatraPDF (PDF) + Chrome headless (pages) + raw ESC/POS
                       Linux/macOS: CUPS lp + Chrome headless
   ▼
Your printer — or ~/.advprint/files if the destination is "save as PDF"
```

- **Hub** = tiny local web server (`advprint.py`) that owns the queue, the history and the printers.
- **Extension** = the browser side: auto-print rules, popup controls, the Options page (tab-wise), page API.
- **Dashboard** = the hub's browser UI at `/dashboard` : status, queue, history, audit, debug, LAN, live test, settings.
- **Tray icon** (Windows) = one click to the dashboard, pause/resume, test page, auto-run toggle.
- **Desktop tool** = `AdvPrint Tool` (Start Menu) — numbered menu for start/stop/service/live test/updates plus **[13] Diagnose & fix** (repairs “hub offline” automatically).

2. Install in 3 minutes

2.1 Windows — the one-time installer (recommended)

1. Download `AdvPrint-Pro-3.0.1-source.zip` and unzip it.
2. Run `AdvPrint-Pro-Setup-3.0.1.exe` . Windows asks for admin once (UAC — the exe requests it itself), no Python needed — the setup carries a **complete Python 3.11.9 runtime** (this is exactly what the broken 2.5.3 download missed: if an installer is smaller than ~9 MB it is incomplete — re-download).
3. Setup registers the **Startup app "AdvPrint Pro"** (Task Manager → Startup apps), creates Start Menu shortcuts, and starts the hub. A tray icon `A` appears next to the clock.
4. Optional ticks: **Run as Windows service** (headless, services.msc) and **Install browser extension files**.

> Windows says *"cannot access / Access is denied"* on a downloaded setup? That is Mark-of-the-Web on a damaged/incomplete download. The 2.6.x setup is complete (~9.6 MB) and the portable `install\install-windows.bat` unblocks every file automatically. If your AV quarantines the hub, whitelist the folder `%LOCALAPPDATA%\AdvPrintPro` — the hub is a plain stdlib Python script, signed-off and open source (MIT).

2.2 Windows — portable zip (no installer)

1. Unzip. Double-click `tool\windows\portable\install\install-windows.bat` . It unblocks files, frees port 9235, uses the bundled runtime, registers auto-run, health-checks the hub.
2. Undo any time: `UNINSTALL-WINDOWS.bat` .

2.3 Linux / Debian / Ubuntu / Mint — the .deb

```
sudo apt install ./advprint-pro_3.0.1_all.deb
advprint-start            # start now
advprint token            # show the current token
advprint livetest         # full A-to-Z self-test
```

Auto-run at login: run `/opt/advprint/install/install-unix.sh` (installs a **systemd --user** service — no root needed; falls back to XDG autostart). Headless server mode: `sudo /opt/advprint/install/install-unix.sh --service`.

2.4 macOS

Run the portable hub: `python3 tool/unix/hub/advprint.py serve`, then `install-unix.sh` (LaunchAgent auto-run). CUPS printers are used through `lp`; page printing uses Chrome/Edge if installed.

2.5 First print in 60 seconds

1. Hub running (tray icon green / `advprint status`).
2. Load the extension: `chrome://extensions` → Developer mode → **Load unpacked** → the `extension` folder (or the one installed to `%LOCALAPPDATA%\AdvPrintPro\extension`).
3. Open the extension Options → Settings → **Detect token** (one click) → **Connect** shows `CONNECTED`.
4. Click **Send test page**. Paper comes out. Done.

3. The token — and the one-click fix

Every request to the hub carries a token. Since v2.0 a **fresh install generates a private random token** and stores it in `%USERPROFILE%\.advprint\config.json` (Linux/macOS `~/.advprint/config.json`). Upgrades of pre-2.0 installs keep working with the old default `PrintHub-2024` until you rotate it — the hub warns you in the dashboard until then. Pre-2.6 installs with a `~/.printhub` folder are **migrated automatically** to `~/.advprint` on first start (old folder kept as backup).

3.1 The one-click fix (v2.6)

Open a stale bookmark like `/dashboard?token=PrintHub-2024` and you get the friendly error page — with a green **“Fix it now — open the dashboard with the current token”** button. One click and you are in. It works because the hub hands the current token only to programs **on the hub PC itself** (loopback-only `GET /token/current`); other PCs on the LAN still see the manual steps only. The same button exists in the extension Options → Settings → **Detect token**.

3.2 Every way to get the current token

Way	How
Tray icon (easiest)	Right-click the **AdvPrint Pro** tray icon → **Open dashboard** — always opens with the right token.
Dashboard error page	Green **Fix it now** button (hub PC only) — or **Copy current token**.
Extension Options	Settings → **Detect token** button fills and saves it automatically.
Command line	`python advprint.py token` (portable) / `advprint token` (deb).
Config file	`"token"` value in `~/.advprint/config.json`.

3.3 Rotating the token

```
python advprint.py token-rotate        # CLI
POST /token/rotate (header auth)       # API
Dashboard → “Rotate token” button
```

After rotating: press **Detect token** in the extension (or paste the new token), and update old dashboard bookmarks. Rotating **invalidates the old token immediately** — including `PrintHub-2024` on upgraded installs.

4. The browser extension — every Options tab explained

Open the extension icon → ⚙ (or right-click the toolbar icon → Options). The Options page is **tab-wise**: `Queue` `History` `Settings` `Debug` `LAN / PCs` `About`. The extension opens on Queue when something is waiting, otherwise on Settings. Every tab auto-refreshes live.

4.1 Queue tab — live FIFO

- Hub state dot: running / PAUSED / offline, pending count, jobs today, hub version.

- Every waiting job row: id, title, printer, kind, status chip + buttons: **Cancel**, **Shift to PC** (print this job on another PC's hub over the LAN), **Print now** (bypasses pause and printer-status hold — the remedy when jobs sit "stuck").
- **Pause / Resume** queue, **Clear** queue, **Refresh**. Auto-refresh every 3 s.

4.2 History tab — last 100 jobs

- Every job with time, title, printer, kind and a colored status chip — **queued** gray, **printed** green, **failed/dead/canceled** red, **held** orange (with the exact held reason, e.g. "printer offline").
- Row actions: **Retry** (failed jobs — optionally on another printer), **Reprint** (printed jobs — reason goes to the audit log), **Cancel**/**Print now** (waiting jobs).
- History auto-refreshes every 8 s — a stuck "queued" row turns "printed" in front of you.
- “Clear” wipes only the browser-side copy; the hub history stays.

4.3 Settings tab — all switches

- **Power switches**: master **AdvPrint enabled** (OFF = nothing prints, gray OFF badge), **Queue mode** (ON = FIFO order + pause; OFF = instant print with auto-queue fallback), **Debug mode**, **Retry when the hub comes back**, UI language (English / বাংলা / auto).
- **Connection**: Hub URL (`http://127.0.0.1:9235`), Token + **Connect** + **Detect token**; buttons to open the dashboard and the audit log.
- **Printing**: default printer (or "File (save as PDF)"), plus **default print options** sent with every job: copies, pages (odd/even/custom), quality (draft…1200 dpi), per-sheet n-up, duplex, layout, scale, margins, paper size — with live capability hints from the selected printer.
- **Auto-print & routing rules**: URL pattern (glob or regex), page-title filter, per-rule printer, action (auto/PDF/page), per-site ON/OFF, auto-print-on-open, delete. First match wins.
- **Behavior**: print every PDF opened in the browser (simple mode), intercept `window.print()` (also Ctrl+P), close tab after auto-print, failure fallback (new tab vs silent), right-click menu, toolbar badge.
- **Hub settings (advanced)** — v2.6: the hub's own switches, editable live (see §9 for the full list): queue & retries, security/lockout, reprint watermark & reason, audit, thermal text engine + ESC/POS codepage, save-as-PDF folder, keep-files, SumatraPDF path.

4.4 Debug tab — WHEN / WHY / WHERE

- One card: hub version, debug mode on/off, worker alive, queue paused, printers ready, instant auto-queue state.
- **Problems found by the hub** table with the exact fix for each finding.
- **Browser error reports**: everything this browser hit (network, token, page JS) while Debug mode was on.
- Buttons: Refresh · Send test report · Copy report (everything for a bug report) · Open /debug JSON · **Run full self-test (A-to-Z)** — runs the hub's live test right here and prints OK/!! lines (v2.6).

4.5 LAN / PCs tab — share & shift

- Shows this PC's LAN URL (`http://192.168.x.x:9235`) when sharing is ON, with copy button.
- **Enable/Disable LAN sharing** (writes `share_lan` ; restart the hub to rebind + open TCP 9235 in the firewall).
- **Other PCs (peers)**: online/offline + version of each configured peer hub, one-click **Shift all waiting jobs**. Or type a LAN URL and shift everything ad hoc.

4.6 About tab

Version + full brand line, update check against GitHub releases, links to the user guide served by the hub (`/docs`), the developer API (`window.advprint`) and the test tools.

4.7 The popup (toolbar icon)

Master switch, queue mode switch, live queue state, pause/resume, last jobs, test page, screenshot-to-printer, save-page-as-PDF, offline banner with retry counter, token warnings with the exact remedy. Hotkey `Alt+P` prints the current tab. Right-click menu prints page / selection / linked PDF / screenshot.

> **v2.7.0 — popup LAN sharing switches:** the toolbar popup now has a **LAN sharing** section with **Share Receive Mode** (this PC accepts jobs from other PCs = hub `share_lan` , saved straight into the hub config) and **Share Send Mode** (this browser may shift jobs to peer PCs — OFF locks every send-to-PC button here). Both turn gray while the hub is offline and exist in English + Bangla. Nothing was removed anywhere.

5. The hub dashboard — every card and button

Open `http://127.0.0.1:9235/dashboard?token=…` (the tray icon does this with one click). All cards refresh live (5–15 s).

Area	What it shows / does
Status cards	Success rate (7 days) · queue size · jobs today · printers ready · hub version.
Action bar	Pause/Clear queue · Docs (user guide) · History CSV · Audit CSV · Diagnostics JSON · Debug · Health JSON · **Rotate token**.
Token warning	Appears while the old default `PrintHub-2024` is active, with a one-click **Rotate now**.
LAN sharing & other PCs	Enable/disable + this PC's LAN URL + printer-status-hold switch + peer PCs with “Shift all waiting jobs”.
Debug card	Debug mode toggle, worker/queue state, problems with fixes, client error reports, copy report.
Printers	Every printer + live status (ready / offline / paper-out / error).
Queue (waiting + printing)	Cancel · Print now · Shift per job.
Job history (v2.7.0)	Full manager: **filter** by status (incl. one-click “Not printed”), **search** title/printer/#id, load 50/100/300/1000, paging, **tick rows** (check / uncheck all) with a live “N selected” counter, **Reprint selected**, **Reprint all non-printed**, **Clear ALL history** (finished jobs only — waiting jobs are never touched), per-row Retry / Reprint.
Audit trail	Latest events: who printed what from where, reprint reasons, token errors.
LAN sharing applies live (v2.8.2)	Toggling **LAN sharing** starts/stops a second listener on this PC’s LAN IP INSIDE the same request — no hub restart, 127.0.0.1 never stops answering, DHCP IP changes are followed automatically. A failed LAN bind is only logged and can never stop local printing.
Firewall one-click + Test LAN (v2.8.2)	LAN card shows **Windows firewall: ALLOWED / BLOCKED** and an **Allow in firewall** button (one UAC prompt; also in the tray menu). **Test LAN** makes the hub call its own LAN URL: PASS + another PC still failing = firewall; FAIL = listener problem (hub.log explains).
Print destination (v2.9.1)	**ANY printer** - one card decides where every job prints: the OS **default printer**, **any installed printer** (picked from the live list), or an upstream **IP printer** at IP:port (**RAW JetDirect 9100** or **IPP**). **Save destination** applies instantly, the **Test** button pings the IP printer (reachability + latency) before saving, and with `dest_mode:"ip"` the queue **holds** jobs while that printer is unreachable (reason shows in the queue row; **Print now** forces). PCs and phones that added *this* hub as their network printer print through exactly this destination - one setting routes everything. `v2.9.1:` the card and ALL dashboard tables are fully responsive - every column, button and option is visible on load, nothing hides inside an inner scrollbar.
Network printer - IPP (v2.9.0)	The hub is a real IPP printer at `http://<lan-ip>:9235/ipp/print` (same port as the hub - no extra firewall rule). Windows 10/11: Add printer → “Select a shared printer by name” → paste the URL (built-in Microsoft IPP Class Driver). Phones: print services that accept a manual IPP address. No token - printer protocol, size-bounded, audited, switchable with `"ipp_printer": false`.
Network printer - RAW 9100 (v2.9.0)	JetDirect listener on **TCP 9100** (`"raw_printer_port"`, 0 = off) while LAN sharing is ON: Windows “Add a printer using TCP/IP address” → this PC’s IP + port 9100 + device type **Generic Network Card** + your real printer’s driver. Windows renders any file and streams the print-ready bytes; the hub prints them. ESC/POS and ZPL apps can open the same socket.
Send portal QR (v2.8.1)	LAN card → **Copy send URL** / **QR code**: phones scan the QR and upload a PDF to the Send portal — it prints on the default printer. The QR is drawn by a QR encoder embedded in the hub (verified bit-exact; no internet used).
One-click invite (v2.8.1)	LAN card → **Copy invite link**: open the link on another PC and click *Add as LAN print peer* — the peer is saved into that PC's config automatically and health-checked. Removable any time from the same LAN card.
DropZone (v2.8.0)	Print WITHOUT the extension: pick a folder; in Word / Excel / ANY app print to “Microsoft Print to PDF” and save the PDF into it — AdvPrint prints it seconds later and moves it to `done\\`. Live on/off (no restart), delete-after-print option, **Open folder** button. Phones/tablets print through the **Send portal** button (`/send`) — token-protected, 100% local.
Restart hub (v2.8.0)	One button in the toolbar: the hub relaunches ITSELF as a new process (queued jobs, settings and history kept) and this page reloads automatically when `/health` answers again. Same restart lives in the tray menu (Restart hub), AdvPrint Tool [14] and `POST /restart`.
Live debug & test tool + Hub settings	**Send test print & watch live** (job → printer verdict), **Release ALL held jobs**, **Full self-test (A-to-Z)**, SumatraPDF path setting. **v2.7.0:** both panels now sit properly INSIDE the page (they used to hang below it, unstyled and full-width).

6. Printing everything

6.1 PDFs (the lab-report flow)

- **Simple mode** (default ON): any PDF opened in the browser prints silently and the tab closes.
- **Rules**: URL pattern + optional title filter + optional printer → auto-prints the moment the URL opens (glob or regex).
- **From your app's code**: `printPdfLink(url, 'Lab report', {fallback:false})` — the page API prints via the hub and never opens a dialog. Full API: extension `content/api.js`, docs at `/docs`.

6.2 Web pages

Right-click → AdvPrint Pro → Print page · `Alt+P` · popup button · `window.advprint.printPage()`. Rendered with headless Chrome/Edge at print quality. `window.print()` and Ctrl+P are intercepted (toggle in Settings) and routed silently.

6.3 Receipts / raw text (ESC/POS)

Send `kind:"raw"` or `|center|…|` markup to `/print`. Codepage selectable (cp437/cp850/cp866/cp1251/cp1252). Bangla and other complex scripts are automatically rendered through headless Chrome so they shape correctly on thermal rolls (`thermal_text_engine`).

6.4 Selections & screenshots

Right-click a selection to print exactly it; the popup can print the visible page as an image — useful for POS systems and stubborn apps.

6.5 Save as PDF instead of paper

Choose printer `File (save as PDF)` or send `"printer":"__file__"`. Files land in `~/Downloads/AdvPrint` or the folder set in Hub settings (`file_dir`).

7. LAN / IP — share one printer with every PC

v3.0.1 — TRAY MENU CLICK FIX + HOOK SWITCH INTO POWER. The tray menu bug where 'the text and the click are not the same' is fixed at the root: the click handler used a selectable-row number as a raw item index, so every item after the first separator ran the WRONG command (Pause queue did nothing, Start at logon paused the queue, Restart hub toggled autostart, Allow LAN in firewall restarted the hub, Exit opened the firewall prompt). Now every item triggers its own action. The Intercept window.print() switch moved into Options > Power switches, next to the master switch (same hookPrint setting as the popup toggle).

v3.0.0 — RENUMBERED RELEASE. The v2.9.1 packages were already distributed, so this release ships the identical v2.9.1 feature set under the clean new number 3.0.0 - no feature added or removed; every surface (hub, dashboard badge, extension, Setup exe, deb, docs) says 3.0.0 and the new packages can never collide with installed 2.9.1 copies. Installs straight over 2.8.2 / 2.9.0 / 2.9.1.

v2.9.1 — RESPONSIVE DASHBOARD + POLISH. The dashboard tables (LAN card, Debug, Printers, Queue, Job history, Audit, **Print destination**) no longer push wide rows into an inner horizontal scrollbar: cells wrap, long titles break, action buttons flow onto the next line - **every column and option is visible the moment the page loads**, on any window width (and on phones the cards go 2-up, controls shrink). The IP / Port / Proto fields of the Print destination card stack cleanly when they appear. Pure CSS/JS change - no settings, no data migration, install over 2.9.0 (or 2.8.2) and refresh.

v2.9.0 — NETWORK PRINTER MODE + PHONE AUTH FIX + SMOOTH TRAY. The hub is now a real IP-printable network printer: add it from another PC with “Select a shared printer by name” → `http://<this-PC-IP>:9235/ipp/print` (Windows installs it with the built-in Microsoft IPP Class Driver) or the classic **“Add a printer using TCP/IP address”** flow — the hub listens on **TCP 9100** (JetDirect RAW) while LAN sharing is ON. PDF, JPEG, PostScript, PCL, ESC/POS and ZPL payloads are all routed to the real printer and every job lands in History + Audit. **Print destination — ANY printer:** the new dashboard card picks where every job prints — the OS **default printer**, **any installed printer**, or an upstream **IP printer** (a real network printer at IP:port, RAW JetDirect or IPP; the **Test** button pings it, and with `dest_mode:"ip"` the queue holds jobs while that printer is unreachable). Phones and PCs that added this hub as their IP printer print through exactly that destination; the LAN card **“IP Printer mode”** row shows the whole chain with a one-click **Turn on**. The phone **“Auth fail” on the Send portal QR is fixed** (the portal upload now accepts the token from the QR link in every token mode). The extension shows **one version everywhere** (read from the manifest), and every tray action runs in its own thread so the menu never lags.

v2.8.2 — LAN SHARING APPLIES LIVE + FIREWALL ONE-CLICK. The reported problem “LAN sharing ON but the LAN IP does not answer” is fixed at the root: no restart needed any more, the firewall state is visible, one click adds the rule, and Test LAN pinpoints the failing side.

v2.8.1 — QR CODE FOR PHONES + ONE-CLICK INVITE (both 100% local).
Send portal (phones) row on the dashboard LAN card: **Copy send URL** copies `http://<this-PC-ip>:9235/send?token=…` and **QR code** shows the same URL as a scannable QR (generated inside the hub — no internet, no CDN). Any phone or tablet on the same Wi-Fi scans it, opens the Send portal, picks a PDF — it prints on this PC's default printer seconds later.
Connect another PC row: **Copy invite link** copies `http://<this-PC-ip>:9235/connect?t=<token>&n=<PC-name>`. Open that link on the OTHER PC (chat, e-mail, QR — any way): it shows a confirmation page with an **Add "PC-name" as a LAN print peer** button. One click saves this hub into the other PC's `peer_hubs` automatically (its own hub does the writing via `POST /peer-connect`), checks the hub is online and shows the peer's version. No retyping IP or token, no config.json editing. Safe by design: loopback-only, private-LAN URLs only, explicit click required, Origin must match the added hub, and every accept is audited (`peer.connected`). API: `GET /qr?token=…&kind=send|invite` returns the QR as SVG.

v2.8.0 — TWO HUB PROFILES (Self PC + LAN PC). The extension now keeps two complete hub profiles. **Connection 1 - Self PC** = this computer's hub URL + token (used when Share Send is OFF — the default). **Connection 2 - LAN PC** = the OTHER PC's hub URL + its token. The popup **Share Send Mode** toggle is the routing switch: **ON → every print job prints on the LAN PC**; OFF → prints on this PC. Recent jobs and Pause/Resume in the popup follow the active target, and Options has four buttons — Open dashboard / audit log — Self PC and — LAN PC. Receiving (Share Receive Mode, hub `share_lan`) is configured on the PC that OWNS the printer; a sender only needs that PC's URL + token. The setup adds the TCP 9235 firewall rule automatically, so "restart the hub and allow the port" is usually the only manual step.

1. On the **printer PC**: extension Options → LAN / PCs → **Enable LAN sharing** (or dashboard → LAN card). Restart the hub.
2. Firewall: allow inbound TCP 9235 (Windows: `netsh advfirewall firewall add rule name="AdvPrint Pro" dir=in action=allow protocol=TCP localport=9235` — run once as admin).
3. On each **client PC**: install the extension, set **Hub URL** to `http://<printer-PC-ip>:9235`, press **Detect token** on the printer PC (or paste the token) → Connect.
4. Every PC now prints to the shared printers; jobs can be **shifted** between PCs (single job or the whole waiting queue) when a PC goes down.

The token is required from every LAN PC — no token, no printing (the hub rate-limits and locks out foreign IPs after too many bad tokens, while your own PC (127.0.0.1) is **never** locked out by default).

8. Queue, History & Audit trail

- **Queue mode ON**: jobs wait in FIFO order, Pause/Resume works, one job prints at a time.
- **Queue mode OFF (instant)**: each job prints immediately; if the printer is not active the job is **auto-queued** (`instant_auto_queue`) and prints when the printer is back.
- **Printer-status hold** (`hold_offline`): jobs whose printer reports offline/paper-out are held with a reason and auto-print on recovery — or hit **Release ALL held jobs** / per-job **Print now** to force.
- **Retries**: failed jobs retry with exponential backoff (`max_retries` × `retry_base_seconds`).
- **Duplicates**: identical jobs inside `dedupe_minutes` are blocked (double-click protection).
- **History — v2.7.0 manager**: last 50/100/300/1000 jobs in the hub DB (SQLite), per-job file for reprints (`keep_files`); in the dashboard AND in Options → History: **filter** by status, **search**, **tick** rows, **Reprint selected**, **Reprint all non-printed**, **Clear ALL history** (finished jobs only — queued/held jobs survive). Export CSV from the dashboard. API: `POST /jobs/bulk-reprint` (`{"ids":[…]}` or `{"scope":"not_printed"}`), `POST /history/clear`.
- **Audit trail**: who printed what, from which site/IP, reprint reasons, auth failures — CSV export too; disable with `audit_enabled:false`.

9. Settings reference — every switch in one table

Change from: extension Options → Hub settings (live) · dashboard · `~/.advprint/config.json` · environment variables (`ADVPRINT_*`; legacy `PRINTHUB_*` still honored).

Key	Default	Meaning
`token`	random per install	API token — rotate via /token/rotate. Old default PrintHub-2024 kept on legacy upgrades.
`token_header_only`	false (true on fresh installs)	Strict mode: POST calls must send the X-AdvPrint-Token header.
`allowed_origins`	[] (all)	Restrict which websites may print, e.g. ["https://www.htncr.org"].
`max_auth_failures / lockout_minutes / lockout_exempt_loopback`	10 / 5 / true	Lockout policy for wrong tokens; loopback exempt by default (owner can never lock himself out).
`printer`	"	Force a printer; empty = OS default; "__file__" = save as PDF.
`printer_groups`	{}	{"Front Desk": ["A","B"]} — fallback printers in order.
`profiles`	{}	{"Invoice": {"paper":"a4","duplex":true}} — named option sets; send "profile":"Invoice".
`queue_mode (extension)`	true	FIFO queue vs instant print.
`max_retries / retry_base_seconds`	2 / 30	Exponential retry of failed jobs.
`dedupe_minutes`	5	Duplicate block window.
`hold_offline`	true	Hold jobs while the printer reports offline/out of paper.
`instant_auto_queue`	true	Instant mode falls back to queue when no printer is active.
`share_lan`	false	Listen on 0.0.0.0 so other PCs can print (token always required).
`peer_hubs / peer_forwarding`	[] / false	Other PCs' hubs for queue shifting / auto-forwarding.
`thermal_text_engine / escpos_codepage`	auto / cp437	Raw text rendering (Bangla → browser engine) and ESC/POS codepage.
`audit_enabled / reprint_watermark / reprint_reason_required`	true / false / false	Audit trail + compliance switches for reprints.
`file_dir / keep_files`	~/Downloads/AdvPrint / true	Save-as-PDF destination; keep files for reprint.
`drop_enabled / drop_dir / drop_delete`	true / "" / false	**v2.8.0 DropZone**: watch a folder and print every PDF that lands in it (default folder `data\DropZone`; drop_delete=true deletes instead of moving to `done\`).
`sumatra_path`	"	Manual SumatraPDF.exe path (empty = auto-find).
`dest_mode / printer`	"" / ""	**v2.9.0 Print destination**: where jobs print — `default` = OS default printer, `selected` = the printer named in `printer` (any installed printer), `ip` = forward to the upstream IP printer. A per-job `printer` always wins.
`ip_printer_host / ip_printer_port / ip_printer_proto`	"" / 0 / raw	**v2.9.0** the upstream IP printer: IP or hostname, port (0 = protocol default: RAW 9100 / IPP 631), protocol `raw` (JetDirect) or `ipp`. Test with `POST /ip-printer/test` or the card's Test button.
`ipp_printer / raw_printer_port`	true / 9100	**v2.9.0 IP printer (inbound)**: this hub IS a network printer — IPP at `/ipp/print` and the JetDirect RAW port for "Add a printer using TCP/IP address".
`debug`	false	Verbose [DEBUG] logs + client error reports in /debug.
`log_max_mb / log_backups`	5 / 3	hub.log rotation.

10. Debug & the Live Test tool

Four layers, from one-click to full control:

1. **Full self-test (A-to-Z)** — v2.6: extension Debug tab button · dashboard “Full self-test” · `AdvPrint Tool` menu [11] · CLI `python advprint.py livetest` (add `print` to also queue a real test page). Checks: Python, data folder writable, token, SumatraPDF/lp, browser, port, queue DB mechanics, printers, default printer — each line `[OK]` or `[!!]` with the fix. Exit code 0 = production-ready.
2. **Test print & watch live** — dashboard button queues a real page and follows the job to the printer with a verdict (REAL PRINTER HIT / SAVED / FAILED + reason). Tray icon → Print test page. CLI `testprint`.

3. **Debug mode** — verbose logs + browser error reports; the Debug tab and `/debug` answer "WHY is nothing printing?" with concrete problems + fixes.
4. **Bundles for support** — `/diagnostics`, `/health`, `/debug` JSON, audit CSV, history CSV, "Copy report" buttons. Token is redacted in diagnostics.

```
$ python advprint.py livetest
  [OK] Python >= 3.8                       3.11.9
  [OK] Data folder + history.db writable   /home/you/.advprint
  [OK] API token configured                private random token
  [OK] CUPS lp command                     /usr/bin/lp
  [OK] Chrome/Edge (page printing)         /usr/bin/chromium
  [OK] Port 9235 available                 OK
  [OK] Queue DB enqueue/read/update        job #42
  [OK] Printers detected                   HP_LaserJet, XP-80C
  [OK] Default printer                     HP_LaserJet

  ALL CHECKS PASSED - AdvPrint Pro is ready for production.
```

11. Enterprise edition — silent install, policy, service

11.1 Silent / scripted Windows install

```
AdvPrint-Pro-Setup-2.9.0.exe /S          # silent (UAC once), hub auto-starts
# enterprise rollout via Intune/GPO/SCCM: run per user, no elevation needed
```

11.2 Managed policy (force settings on all PCs)

Deploy `extension/managed.json` content as Chrome policy `HKLM\SOFTWARE\Policies\Google\Chrome\ExtensionInstallForcelist` + managed storage, or simply push the keys. Forceable settings: `hubUrl`, `token`, `defaultPrinter`, `enabled`, `queueMode`, `retryOfflin e`, `lang`, `hookPrint`, `autoPrintAllPdfs`, `failFallback`, `closeAfterPrint`, `rules` (rules fully override local ones — central lockdown). Managed values win over user settings.

11.3 Windows service mode (headless)

```
python advprint.py service-install     # UAC → registers services.msc: "AdvPrint Pro Service"
python advprint.py service-status / service-remove
```

Service mode = no tray, runs before logon, perfect for a print-server PC. The normal Startup app stays the default on desktops; if both run, the second exits gracefully (port guard).

11.4 Lockdown checklist

- `allowed_origins` → only your web app may print.
- `token_header_only:true` + rotated token on every PC.
- `share_lan:false` everywhere except the print-server PC; that one keeps `lockout_exempt_loopback` and strict lockouts for foreign IPs.
- `audit_enabled:true` + `reprint_reason_required:true` for compliance; audit CSV ships out nightly.
- LAN firewall rule only where needed (§7).

12. HTTP API quick reference

Auth: header `X-AdvPrint-Token` (legacy `X-PrintHub-Token` accepted), or `?token=` on GET navigation. Full docs: `docs/API.md` and the hub's own `/docs` page.

```
# v2.9.x - NETWORK PRINTER MODE: print INTO this hub from any device
#   IPP (Windows "Select a shared printer by name", phones):
curl -X POST -H "Content-Type: application/pdf" --data-binary @report.pdf \
     http://:9235/ipp/print            # no token - printer protocol
#   RAW JetDirect: anything streamed to TCP 9100 prints the same way.
# Print destination - route every job to an upstream IP printer:
curl -X POST -H "X-AdvPrint-Token: $TOKEN" -H "Content-Type: application/json" \
  -d '{"dest_mode":"ip","ip_printer_host":"192.168.1.50",
       "ip_printer_port":9100,"ip_printer_proto":"raw"}' \
  http://127.0.0.1:9235/config
curl -X POST -H "X-AdvPrint-Token: $TOKEN" \
     "http://127.0.0.1:9235/ip-printer/test?host=192.168.1.50&port=9100&proto=raw"

# v2.8.0 - DropZone / send portal / restart (no extension needed)
curl -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/dropzone
curl -X POST -H "X-AdvPrint-Token: $TOKEN" -F "file=@report.pdf" \
     "http://127.0.0.1:9235/send?title=Report"   # upload + print
curl -X POST -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/restart

# status
curl http://127.0.0.1:9235/health
curl -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/printers
# print a PDF from bytes (silent)
curl -X POST -H "X-AdvPrint-Token: $TOKEN" -H "Content-Type: application/pdf" \
     --data-binary @report.pdf "http://127.0.0.1:9235/print?title=Report"
# print a page with options
curl -X POST -H "X-AdvPrint-Token: $TOKEN" -H "Content-Type: application/json" \
  -d '{"url":"https://intranet/roster","paper":"a4","duplex":"duplex","copies":2}' \
  http://127.0.0.1:9235/print
# queue control / tokens / settings
curl -X POST -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/queue/pause
curl -X POST -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/token/rotate
curl -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/config
# live test over HTTP (same check as the CLI)
curl -X POST -H "X-AdvPrint-Token: $TOKEN" http://127.0.0.1:9235/livetest
```

Page-side API (in any web page once the extension is installed): `printPdfLink(url, title, opts)` · `window.advprint.printPdf/printPage/printers/status/history`.

13. Troubleshooting — every known problem and its fix

Symptom	Cause → Fix
Setup says *cannot access / Access is denied*	Incomplete/damaged download (pre-2.6 bug) or Mark-of-the-Web → re-download the 2.8.1 zip (setup ≈ 9.7 MB), right-click zip → Properties → Unblock, or run `install-windows.bat` which unblocks automatically.
Dashboard shows “needs the current token”	Stale bookmark → press the green **Fix it now** button on that very page (hub PC), or tray → Open dashboard, or extension → Detect token.
Extension shows 401 / token rejected	Options → Settings → **Detect token** (or paste the token from `advprint token`). After a rotation always re-detect.
Nothing prints, no error	Debug tab → Run full self-test → fix the [!!] line (usually SumatraPDF missing or no default printer). Then send a test print & watch live.
Jobs stuck “queued”	Queue paused → Resume; printer held → **Print now** or Release ALL held jobs; check printer status in the dashboard Printers card.
Jobs stuck “held”	That is the printer-status hold doing its job — fix the printer or **Print now**; disable hold in Hub settings (`hold_offline`).
Bangla/receipt text is garbage	Hub settings → thermal engine `auto` (browser render) + correct codepage; browser printing needs Chrome/Edge installed.
LAN PC cannot connect	`share_lan` ON + hub restarted + firewall TCP 9235 + correct Hub URL + token; check the LAN/PCs tab shows the PC's URL.
“python” in Task Manager	You started with system python — the installer's `AdvPrintPro.exe` (renamed pythonw) shows “AdvPrintPro”; re-run setup or the Startup shortcut.
Auto-run missing	Task Manager → Startup apps → “AdvPrint Pro” enable; or tray menu → Start at Windows logon.
Old PrintHub Pro traces after upgrade	Setup 2.6 removes the old Startup app/task/service/registry keys automatically; leftovers: uninstall the old version once, or delete `%LOCALAPPDATA%\PrintHubPro`.
Hub shows OFFLINE / start “did not answer”	Open the desktop tool → **[13] Diagnose & fix**: it unblocks the folder (Mark-of-the-Web), frees port 9235 from zombies, then starts the hub in plain view so the real error shows as text. Ends with the hub.log tail + a copy/paste start command with the full path.
Tool prints the diagnosis but hub.log is empty	The hub died before logging (AV quarantined the fresh exe) → add the AdvPrint folder to the antivirus exclusions, run [13] again (it unblocks), then [1] Start.
“python advprint.py serve” says *can't open file*	You are in the wrong folder — that command needs the file's real location. Use the tool's copy/paste command (menu [13], step [5/5]) or just menu [1].
Dashboard tables show “-” and buttons do nothing	Old-build JS bug (one broken quote killed the whole dashboard script) — **fixed in 2.6.4**; update, hard-refresh the dashboard (Ctrl+F5).
Tray icon missing	Fixed in 2.6.4 (embedded-runtime isolated mode). Update + start the hub; the icon sits next to the clock — may hide under the `^` arrow. If the hub is not running there is no tray — start with [1] first.
Hub will not start (port busy)	Another instance runs — check the tray icon; the hub exits gracefully with a clear message instead of crashing; [13] can free the port for you.
Screen “glitch” / black console flashes every few seconds	Old build (pre-2.6.6): the hub polled the service state with `sc query` without the windowless flag — Windows flashed a console each time. **Fixed in 2.6.6** — every background child now runs windowless; update.
`save-as-PDF failed: [Errno 13] Permission denied`	A real-time antivirus / indexer briefly locked the fresh PDF. **Fixed in 2.6.7** — the hub now writes atomically and retry-reads ~4 s. If it still repeats: add `C:\Users\<you>\.advprint` to your AV exclusions, then **Retry** the job from History.
`print backend exit 1:` with no message (File (PDF))	**Fixed in 2.6.7** — “File (PDF)” is now a virtual printer: the PDF is **saved** to `Downloads\AdvPrint` with status “saved” (also with Queue Mode OFF). The backend's stdout is captured too, so real printer errors now show a readable reason.
Windows service “AdvPrint” stuck in *start pending*	A stale state left by an older service process — the hub actually runs from the Startup app / tray. Fix: `services.msc` → AdvPrint → Restart (or reboot), or run `service-remove` — keep only ONE of service / Startup app active. The Debug tab lists this automatically.

`printer 'X...X not found` — the name shows TWICE in a row (e.g. `CanonCanon`)	A stored printer name was typed/pasted twice. **Self-healed in 2.6.8** — the hub collapses the doubled name to the real printer, prints, and corrects the history (audit: `printer.corrected`). Old dead jobs: just press **Retry**. Still check Settings → Printer for the stale entry.
Every page prints TWICE — two sheets, and the log shows two `[OK] job #N` lines	Queue Mode OFF race in v2.6.8 and earlier: the FIFO worker and the direct thread both grabbed the same job. **Fixed in 2.6.9** — the job row is created in its final state, one job = one sheet. Not a printer or driver problem (the printer was innocent). Upgrade the hub to 2.6.9+; nothing to change on the printer.
Right-click "Print this page" fails with *Cannot access contents of url chrome-extension://…*	The menu was used on a browser-internal page (chrome://, settings, extension page). **Fixed in 2.6.8** — the menu now only appears on printable pages (http/https/file) and shows a friendly notice otherwise.
Phone shows *invalid or missing token* (“Auth fail”) on the Send portal	Old builds accepted the token header only, which a phone browser cannot send. **Fixed in 2.9.0** - the /send upload accepts the token from the QR link itself in every token mode; re-open the QR URL (regenerate it from the LAN card if the token was rotated).
Jobs show *held - IP printer x.x.x.x:9100 is unreachable*	`dest_mode:"ip"` and the Test/probe says the upstream printer is down: fix the printer (power/cable/IP change), then **Print now** or Release ALL held jobs. Check the address with the card's **Test** button first.
Dashboard tables/buttons are cut off, must scroll inside the card to see all options	**Fixed in 2.9.1** - tables wrap instead of scrolling; update the hub and hard-refresh the dashboard (Ctrl+F5).
HP Laser 103/107/108 shows *offline*	That is the printer device itself (USB/network/sleep), not a hub error — check cable/power, set it online in Windows, then **Retry** the held jobs.

14. Upgrade, uninstall, files & folders

v2.8.0 — BULLET-PROOF UPGRADE & UNINSTALL. Setup runs `hub-stop.ps1` BEFORE touching files: it stops the optional *AdvPrint* service (wait + PID kill when stuck), **deletes the service registration** (it is reinstalled fresh automatically on upgrade), kills every process under the install folder and frees TCP 9235. Old program folders are wiped first, so no stale file can abort the new install any more. The uninstaller uses the same stop, removes the firewall rule and retries the folder sweep 3×. Silent install: `AdvPrint-Pro-Setup-2.8.0.exe /S` (UAC once — the exe requests admin itself; the hub auto-starts afterwards).

14.1 Upgrade

Run the new setup over the old one (same per-user location, settings/history kept) — or replace the portable folder. The extension updates by Remove + Load unpacked (or GitHub update check → new zip). Data folders migrate: `~/.printhub` → `~/.advprint` automatically.

14.2 Uninstall

Windows: Start Menu → Uninstall AdvPrint Pro (or Apps & features). Service: removed automatically. History/settings in `~/.advprint` are kept (delete manually if unwanted). Linux: `sudo apt remove advprint-pro` + `uninstall-unix.sh`.

14.3 Where everything lives

What	Path
Program (Windows)	`%LOCALAPPDATA%\AdvPrintPro` (hub, runtime, install scripts, extension, docs)
Data (config, history.db, hub.log, files)	`%USERPROFILE%\.advprint` (migrated from `.printhub`)
Auto-run	Startup app “AdvPrint Pro” (Task Manager) · optional service “AdvPrint” (services.msc)
Program (Linux deb)	`/opt/advprint` + `/usr/bin/advprint*` · data `~/.advprint`
Logs	`~/.advprint/hub.log` (rotating)

You are never locked out: the hub never locks out your own PC (127.0.0.1) by default, the tray always opens the dashboard with the right token, and the error page itself offers the one-click fix. That is the v2.6 promise: **auto security, zero manual Security-tab surgery**.

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